HW 08

P1) Jackson 2.2.

a) If we put an image charge q' at distance $|\vec{y}'|$ outside the sphere so that

$$q' = -\frac{a}{y} q,$$

$$y' = \frac{a^2}{y}$$

the potential $\Phi(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\left[\frac{q}{|\vec{r}-\vec{y}|} + \frac{q'}{|\vec{r}-\vec{y}'|}\right]$

goes to zero on the sphere. Thus the potential inside the sphere is

$$\Phi(\vec{r}) = \frac{q}{4\pi\varepsilon_0}\left[\frac{1}{|\vec{r}-\vec{y}|} - \frac{1}{|\frac{y}{a}\vec{r} - \frac{a}{y}\vec{y}|}\right].$$

b) The normal derivative is radially inward to the sphere. Therefore there is an additional minus sign:

$$\sigma = -\varepsilon_0 \frac{\partial \Phi}{\partial(-x)}\Big|_{x=a} = \frac{q}{4\pi a y} \frac{1 - a^2/y^2}{(1 + a^2/y^2 - 2a/y \cdot \cos\theta_{\vec{r},\vec{y}})^{3/2}}$$

c) The force is directed to the image charge q'. Its absolute value is:

$$F = \frac{q^2}{4\pi\varepsilon_0} \cdot \frac{a y}{(a^2 - y^2)^2}$$

d) There are some changes in the solution; in both cases we have to add some constants to the potential.

1) When the sphere is kept at a fixed potential V, the total potential inside is the sum of two terms: the potential of a grounded sphere and the constant potential V. This is because the potential must be continuous, and there should be no field inside the sphere without additional charges. So the potential in this case is

$$\Phi(\vec{r}) = \Phi(\vec{r})\Big|_{V=0} + V$$

2) If the sphere has a total charge Q, then there is a charge $Q+q$ at the outer surface of it. Therefore the potential outside the sphere is

$$\Phi_{out} = \frac{1}{4\pi\varepsilon_0}\frac{Q+q}{r},$$

which tells us that the potential at the sphere is $V' = \frac{1}{4\pi\varepsilon_0}\frac{Q+q}{a}$. Now from the previous solution we conclude that

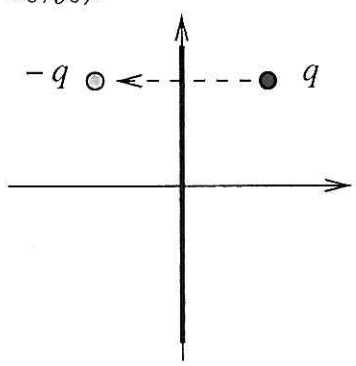
$$\Phi(\vec{r}) = \Phi(\vec{r})\Big|_{Q=0} + \frac{1}{4\pi\varepsilon_0}\frac{Q+q}{a}$$

P2

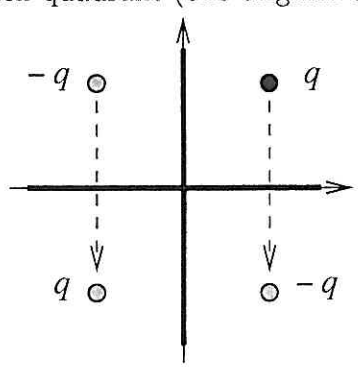
2.3 A straight-line charge with constant linear charge density λ is located perpendicular to the x-y plane in the first quadrant at (x_0, y_0). The intersecting planes $x = 0$, $y \geq 0$ and $y = 0$, $x \geq 0$ are conducting boundary surfaces held at zero potential. Consider the potential, fields, and surface charges in the first quadrant.

a) The well-known potential for an isolated line charge at (x_0, y_0) is $\Phi(x, y) = (\lambda/4\pi\epsilon_0)\ln(R^2/r^2)$, where $r^2 = (x - x_0)^2 + (y - y_0)^2$ and R is a constant. Determine the expression for the potential of the line charge in the presence of the intersecting planes. Verify explicitly that the potential and the tangential electric field vanish on the boundary surfaces.

The image charges for this solution may be built up one plane at a time. If there were only a plane at $x = 0$, then the line charge λ at (x_0, y_0) will generate an image charge $-\lambda$ at $(-x_0, y_0)$.



Adding a plane at $y = 0$ will then result in additional image charges for the above two charges: an image of the original charge giving $-\lambda$ at $(x_0, -y_0)$ and an image of the initial image giving λ at $(-x_0, -y_0)$. The final picture is that of four charges, one in each quadrant (one original and three images).



Adding up the contributions of these four charges yields

$$\begin{aligned}\Phi(x,y) &= \frac{\lambda}{4\pi\epsilon_0}\left(\ln\frac{R^2}{(x-x_0)^2+(y-y_0)^2}+\ln\frac{R^2}{(x+x_0)^2+(y+y_0)^2}\right.\\ &\qquad\left.-\ln\frac{R^2}{(x-x_0)^2+(y+y_0)^2}-\ln\frac{R^2}{(x+x_0)^2+(y-y_0)^2}\right)\\ &= \frac{\lambda}{4\pi\epsilon_0}\ln\frac{[(x-x_0)^2+(y+y_0)^2][(x+x_0)^2+(y-y_0)^2]}{[(x-x_0)^2+(y-y_0)^2][(x+x_0)^2+(y+y_0)^2]}\end{aligned} \tag{1}$$

Note that the arbitrary constant R drops out of this expression.

We may check the validity of this solution by evaluating the potential on the boundary surface $x=0$ ($y=0$ is obviously similar)

$$\Phi(0,y) = \frac{\lambda}{4\pi\epsilon_0}\ln\frac{[x_0^2+(y+y_0)^2][x_0^2+(y-y_0)^2]}{[x_0^2+(y-y_0)^2][x_0^2+(y+y_0)^2]} = \frac{\lambda}{4\pi\epsilon_0}\ln 1 = 0$$

For the tangential electric field, we may compute E_y and examine its behavior on the surface $x=0$. Before setting $x=0$, however, we find

$$\begin{aligned}E_y(x,y) = -\partial_y\Phi(x,y) &= \frac{\lambda}{2\pi\epsilon_0}\left(\frac{y-y_0}{(x-x_0)^2+(y-y_0)^2}+\frac{y+y_0}{(x+x_0)^2+(y+y_0)^2}\right.\\ &\qquad\left.-\frac{y+y_0}{(x-x_0)^2+(y+y_0)^2}-\frac{y-y_0}{(x+x_0)^2+(y-y_0)^2}\right)\end{aligned} \tag{2}$$

Finally, setting $x=0$ yields

$$\begin{aligned}E_y(0,y) &= \frac{\lambda}{2\pi\epsilon_0}\left(\frac{y-y_0}{x_0^2+(y-y_0)^2}+\frac{y+y_0}{x_0^2+(y+y_0)^2}\right.\\ &\qquad\left.-\frac{y+y_0}{x_0^2+(y+y_0)^2}-\frac{y-y_0}{x_0^2+(y-y_0)^2}\right)=0\end{aligned}$$

The result is similar for the $y=0$ surface.

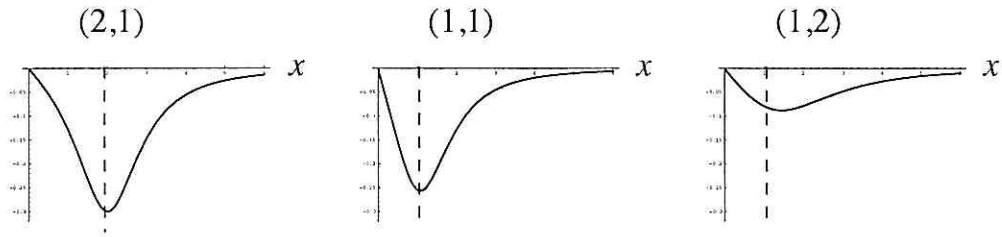
b) Determine the surface charge density σ on the plane $y=0$, $x\geq 0$. Plot σ/λ versus x for $(x_0=2, y_0=1)$, $(x_0=1, y_0=1)$, and $(x_0=1, y_0=2)$.

The surface charge density on the $y=0$ plane is obtained from the normal component of the electric field, $\sigma=\epsilon_0 E_y(x,0)$. Setting $y=0$ in (2) gives

$$\begin{aligned}\sigma &= \frac{\lambda}{2\pi}\left(\frac{-y_0}{(x-x_0)^2+y_0^2}+\frac{y_0}{(x+x_0)^2+y_0^2}-\frac{y_0}{(x-x_0)^2+y_0^2}-\frac{-y_0}{(x+x_0)^2+y_0^2}\right)\\ &= \frac{\lambda y_0}{\pi}\left(\frac{1}{(x+x_0)^2+y_0^2}-\frac{1}{(x-x_0)^2+y_0^2}\right)\\ &= -\frac{4\lambda x_0 y_0}{\pi}\frac{x}{[(x-x_0)^2+y_0^2][(x+x_0)^2+y_0^2]}\end{aligned} \tag{3}$$

The expression in the last line demonstrates that the surface charge density vanishes (linearly with x) at the origin. This is an explicit verification of the result for charge density at the boundary of two parallel conductors making a 90° angle.

The plots of σ/λ are as follows



Note that, at a fixed x_0 position ($x_0 = 1$) the magnitude of the induced charge density is greater when the charge is closer to the conductor ($y_0 = 1$ instead of 2). This is of course expected. The magnitude is reduced, however, as the charge is brought closer to the second conductor ($x_0 = 1$ instead of 2 for fixed $y_0 = 1$). The reason for this is that in this case more of the induced charge gets spread into the second conductor.

c) Show that the total charge (per unit length in z) on the plane $y = 0$, $x \geq 0$ is

$$Q_x = -\frac{2}{\pi}\lambda \tan^{-1}\left(\frac{x_0}{y_0}\right)$$

What is the total charge on the plane $x = 0$?

The total charge is obtained by integrating the surface charge density (3)

$$\begin{aligned} Q_x = \int_0^\infty \sigma(x)\,dx &= \frac{\lambda y_0}{\pi}\int_0^\infty \left(\frac{1}{(x+x_0)^2+y_0^2} - \frac{1}{(x-x_0)^2+y_0^2}\right) dx \\ &= \frac{\lambda}{\pi}\left[\tan^{-1}\frac{x+x_0}{y_0} - \tan^{-1}\frac{x-x_0}{y_0}\right]_0^\infty = -\frac{2\lambda}{\pi}\tan^{-1}\frac{x_0}{y_0} \end{aligned}$$

The total charge Q_y on the plane $x = 0$ is given by $x \to y$ interchange symmetry

$$Q_y = -\frac{2\lambda}{\pi}\tan^{-1}\frac{y_0}{x_0}$$

Hence the combined total charge is

$$Q_{\text{tot}} = Q_x + Q_y = -\frac{2\lambda}{\pi}\left(\tan^{-1}\frac{x_0}{y_0} + \tan^{-1}\frac{y_0}{x_0}\right) = -\lambda$$

This is of course equivalent to the sum of the charges of the three image charges.

d) Show that far from the origin [$\rho \gg \rho_0$, where $\rho = \sqrt{(x^2+y^2)}$ and $\rho_0 = \sqrt{(x_0^2+y_0^2)}$] the leading term in the potential is

$$\Phi \to \Phi_{\text{asym}} = \frac{4\lambda}{\pi\epsilon_0}\frac{(x_0y_0)(xy)}{\rho^4}$$

Interpret.

Here we simply need to Taylor expand the expression in (1). We start by expanding the squares and dividing out by ρ^2 to obtain

$$\Phi = \frac{\lambda}{4\pi\epsilon_0} \ln \frac{[1 - \frac{2xx_0}{\rho^2} + \frac{2yy_0}{\rho^2} + \frac{\rho_0^2}{\rho^2}][1 + \frac{2xx_0}{\rho^2} - \frac{2yy_0}{\rho^2} + \frac{\rho_0^2}{\rho^2}]}{[1 - \frac{2xx_0}{\rho^2} - \frac{2yy_0}{\rho^2} + \frac{\rho_0^2}{\rho^2}][1 + \frac{2xx_0}{\rho^2} + \frac{2yy_0}{\rho^2} + \frac{\rho_0^2}{\rho^2}]}$$

Assuming x and y are both of $\mathcal{O}(\rho)$ and x_0 and y_0 are of $\mathcal{O}(\rho_0)$, we see that xx_0/ρ^2 and yy_0/ρ^2 are of $\mathcal{O}(\rho_0/rho)$. This lets us keep track of the expansion in orders of ρ_0/rho. Expanding up to second order gives

$$\begin{aligned}\Phi &= \frac{\lambda}{4\pi\epsilon_0} \ln \frac{1 + \frac{2\rho_0^2}{\rho^2} - \frac{4x^2x_0^2}{\rho^4} - \frac{4y^2y_0^2}{\rho^4} + \frac{8xyx_0y_0}{\rho^4} + \mathcal{O}(\frac{\rho_0^4}{\rho^4})}{1 + \frac{2\rho_0^2}{\rho^2} - \frac{4x^2x_0^2}{\rho^4} - \frac{4y^2y_0^2}{\rho^4} - \frac{8xyx_0y_0}{\rho^4} + \mathcal{O}(\frac{\rho_0^4}{\rho^4})} \\ &= \frac{\lambda}{4\pi\epsilon_0} \ln\left(1 + \frac{16xyx_0y_0}{\rho^4} + \cdots\right) \approx \frac{4\lambda xyx_0y_0}{\pi\epsilon_0\rho^4}\end{aligned}$$

This is basically a two-dimensional quadrupole potential, which is not too surprising as the space is divided into four quadrants (with alternating positive and negative (image) charges). Going to polar coordinates $(x, y) = \rho(\cos\varphi, \sin\varphi)$, we have

$$\Phi \approx \frac{2\lambda x_0y_0}{\pi\epsilon_0} \frac{\sin 2\varphi}{\rho^2}$$

Note that $4\lambda x_0y_0$ is essentially the quadrupole moment (charge times area) and that the $\sin 2\varphi$ angular behavior is obviously quadrupolar (angular momentum $l = 2$).

P3

2.13 Two Cylinder Halves at Constant Potentials

a. The Potential inside the Cylinder

In this case (see figure 2.8) there are no free charges in the area of interest. Therefore the potential Φ inside the cylinder obeys Laplace's equation:

$$\nabla^2\Phi = 0 \tag{2.46}$$

We can write the potential in cylindrical coordinates and separate the variables:

$$\Phi(\rho, \phi) = R(r)F(\phi) \tag{2.47}$$

The general solution is (see J2.71):

$$\Phi(\rho,\phi) \quad = \quad a_0 + b_0 \ln\rho + \sum_{n=1}^{\infty}\left[a_n\rho^n\sin(n\phi+\alpha_n) + b_n\rho^{-n}\cos(n\phi+\alpha_n)\right] \tag{2.48}$$

From this geometry it is obvious that at the center $\rho = 0$ the solution may not blow up, so:

$$b_n = b_0 = 0 \tag{2.49}$$

This results in a potential

$$\Phi(\rho,\phi) \quad = \quad a_0 + \sum_{n=1}^{\infty} a_n\rho^n\sin(n\phi+\alpha_n) \tag{2.50}$$

The next step is to implement the boundary conditions

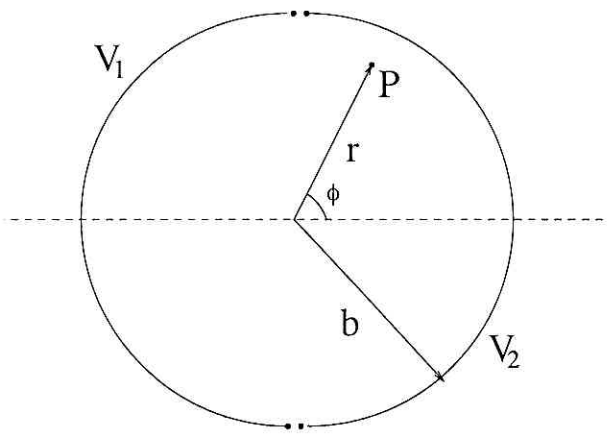


Figure 2.8: Cross-section of two cylinder halves with radius b at constant potentials V_1 and V_2.

$$\Phi(b,\phi) = V_2 = a_0 + \sum_{n=1}^{\infty} a_n b^n \sin(n\phi+\alpha_n) \quad \text{for } (-\pi/2 < \phi < \pi/2)$$
$$\Phi(b,\phi) = V_1 = a_0 + \sum_{n=1}^{\infty} a_n b^n \sin(n\phi+\alpha_n) \quad \text{for } (\pi/2 < \phi < 3\pi/2) \tag{2.51}$$

b. The Surface Charge Density

$$\sigma = -\,\epsilon_0 \frac{\partial\Phi}{\partial r}\bigg|_{r=b} \tag{2.52}$$